

Determinant Quantum Monte Carlo for the Hubbard model

A self-contained derivation of the BSS algorithm and its implementation in `qmc_dqmc`

`qmc_dqmc` contributors

Contents

1	Overview	1
2	Model and conventions	1
3	Trotter decomposition and HS transformation	1
4	Equal-time Green's function	2
5	Single-spin-flip update	2
6	Wrapping between time slices	3
7	Numerical stabilization	3
8	Observables	3
9	Verification	4

1 Overview

The Determinant Quantum Monte Carlo (DQMC) algorithm of Blankenbecler, Scalapino and Sugar [1], equipped with the Hirsch [2] discrete Hubbard–Stratonovich transformation, is the canonical tool for unbiased finite-temperature simulations of the repulsive Hubbard model. This document derives the algorithm implemented in `qmc_dqmc`, focusing on the practical equations needed to read the code in `include/qmc/dqmc_engine.hpp`.

The companion document `sse_qmc.tex` treats the SSE algorithm for spin systems; the present notes are independent and use a similar notation. See also the implementation summary in `docs/DQMC.md`.

2 Model and conventions

We simulate the single-band Hubbard model on a generic lattice with periodic boundary conditions:

$$\hat{H} = -t \sum_{\langle ij \rangle, \sigma} \left(c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.} \right) - \mu \sum_{i\sigma} n_{i\sigma} + U \sum_i \left(n_{i\uparrow} - \frac{1}{2} \right) \left(n_{i\downarrow} - \frac{1}{2} \right). \quad (1)$$

The symmetric $(n - \frac{1}{2})$ form makes $\mu = 0$ coincide with half-filling on bipartite lattices by particle-hole symmetry. We collect the kinetic and chemical-potential terms into a single-particle matrix $K \in \mathbb{R}^{N_s \times N_s}$ with $K_{ij} = -t$ for nearest neighbours and $K_{ii} = -\mu$.

3 Trotter decomposition and HS transformation

Discretise imaginary time as $\beta = L_\tau \Delta\tau$. To leading order in $\Delta\tau$,

$$e^{-\beta\hat{H}} = \prod_{l=0}^{L_\tau-1} e^{-\Delta\tau K} e^{-\Delta\tau V} + \mathcal{O}(\Delta\tau^2), \quad (2)$$

where $V = U \sum_i (n_{i\uparrow} - \frac{1}{2})(n_{i\downarrow} - \frac{1}{2})$. Hirsch's discrete HS identity decouples the on-site quartic term:

$$e^{-\Delta\tau U(n_{i\uparrow} - \frac{1}{2})(n_{i\downarrow} - \frac{1}{2})} = \frac{e^{-\Delta\tau U/4}}{2} \sum_{s=\pm 1} e^{\alpha s(n_{i\uparrow} - n_{i\downarrow})}, \quad \cosh \alpha = e^{\Delta\tau U/2}. \quad (3)$$

Introducing one Hubbard-Stratonovich field $s_{l,i} \in \{-1, +1\}$ at every site of every Trotter slice, we obtain

$$Z = \text{Tr} e^{-\beta\hat{H}} = \frac{e^{-\beta U N_s/4}}{2^{N_s L_\tau}} \sum_{\{s_{l,i}\}} \det M_\uparrow(s) \det M_\downarrow(s), \quad (4)$$

with the single-particle propagation matrices

$$M_\sigma(s) = \mathbb{K} + B_{L_\tau-1,\sigma} B_{L_\tau-2,\sigma} \cdots B_{0,\sigma}, \quad B_{l,\sigma}(s) = e^{-\Delta\tau K} e^{V_{l,\sigma}(s)}, \quad (5)$$

where $V_{l,\sigma}(s) = \alpha \sigma \text{diag}(s_{l,1}, \dots, s_{l,N_s})$ (here $\sigma = \pm 1$ for $\uparrow, \downarrow$). The B matrices are $N_s \times N_s$ and are computed lazily, never stored, in the implementation.

Sign-free regime. On bipartite lattices at $\mu = 0$ particle-hole symmetry maps $c_{i\sigma} \mapsto (-1)^{a_i} c_{i\sigma}^\dagger$ (with a_i the sublattice label) and exchanges the two spin sectors, giving $\det M_\uparrow(s) = \det M_\downarrow(s)$ for every aux-field configuration. The product $\det M_\uparrow \det M_\downarrow \geq 0$, so the Monte Carlo weight is non-negative and the sign problem is absent. Away from half-filling the sign decays as $\langle \text{sign} \rangle \sim e^{-\beta U f(\rho, T)}$ and rapidly becomes the limiting factor in low-temperature simulations of the doped Hubbard model.

4 Equal-time Green's function

The fundamental object measured by DQMC is the equal-time Green's function on time slice l ,

$$G_\sigma(l)_{ij} = \langle c_{i\sigma} c_{j\sigma}^\dagger \rangle_l = [\mathbb{K} + A_\sigma(l)]_{ij}^{-1}, \quad (6)$$

where the cyclic ordering of the propagator product is the *convention* that determines the form of the local-update acceptance ratio. We adopt the *pre- B_l* convention,

$$A_\sigma(l) = B_{l-1,\sigma} B_{l-2,\sigma} \cdots B_{0,\sigma} B_{L_\tau-1,\sigma} \cdots B_{l+1,\sigma} B_{l,\sigma}, \quad (7)$$

which leads (Sec. 5) to the simple ratio $r = 1 + \Delta(1 - G_{ii})$. All Wick-theorem-based observables are expressed below in the same convention.

5 Single-spin-flip update

A flip $s_{l,i} \mapsto -s_{l,i}$ modifies only $B_{l,\sigma}$: $B_{l,\sigma} \rightarrow B_{l,\sigma}(\mathbb{K} + \Delta_\sigma E_{ii})$ where E_{ii} is the rank-one matrix with a single 1 at position (i, i) and

$$\Delta_\uparrow = e^{-2\alpha s_{l,i}} - 1, \quad \Delta_\downarrow = e^{+2\alpha s_{l,i}} - 1. \quad (8)$$

In the pre- B_l convention, this multiplies A on the *right*: $A' = A(\mathbb{K} + \Delta E_{ii})$. A single Sherman–Morrison identity then gives the determinant ratio

$$R_\sigma \equiv \frac{\det(\mathbb{K} + A'_\sigma)}{\det(\mathbb{K} + A_\sigma)} = 1 + \Delta_\sigma (1 - G_\sigma(i, i)). \quad (9)$$

The Metropolis acceptance probability is $p = \min(1, |R_\uparrow R_\downarrow|)$.

If the flip is accepted, the equal-time Green’s function is updated exactly by another Sherman–Morrison step. Setting $u_j = \delta_{ji} - G_\sigma(j, i)$ and noting that $v = e_i$,

$$G'_\sigma = \left(\mathbb{K} - \frac{u e_i^\top}{R_\sigma} \right) G_\sigma, \quad G'_\sigma(j, k) = G_\sigma(j, k) - \frac{\Delta_\sigma}{R_\sigma} (\delta_{ji} - G_\sigma(j, i)) G_\sigma(i, k). \quad (10)$$

This is an $O(N_s^2)$ outer product per accepted move (and $O(N_s)$ per proposal for the ratio).

6 Wrapping between time slices

After all single-spin updates at slice l we advance to slice $l + 1$. From the cyclic ordering of $A(l)$ one gets directly

$$A_\sigma(l + 1) = B_{l,\sigma} A_\sigma(l) B_{l,\sigma}^{-1}, \quad G_\sigma(l + 1) = B_{l,\sigma} G_\sigma(l) B_{l,\sigma}^{-1}. \quad (11)$$

With $B_{l,\sigma} = e^{-\Delta\tau K} \cdot \text{diag}(d)$ and $B_{l,\sigma}^{-1} = \text{diag}(d^{-1}) \cdot e^{+\Delta\tau K}$, the order of operations matters because $\text{diag}(d^{-1})$ does *not* commute with $e^{+\Delta\tau K}$:

$$G_\sigma(l + 1) = e^{-\Delta\tau K} [\text{diag}(d) \cdot G_\sigma(l) \cdot \text{diag}(d^{-1})] e^{+\Delta\tau K}. \quad (12)$$

Equation (12) is what the code in `wrap_green_up_` and `wrap_green_dn_` actually computes; getting this ordering wrong is a subtle source of physical error that *vanishes identically* at $U = 0$, so the free-fermion regression test does not catch it.

7 Numerical stabilization

The matrix product $A_\sigma(l) = B_{l-1,\sigma} \cdots B_{l,\sigma}$ is catastrophically ill-conditioned at low temperatures: each $e^{-\Delta\tau K}$ factor stretches some directions by $e^{+\Delta\tau \lambda_{\max}(K)}$ and contracts others by $e^{-\Delta\tau \lambda_{\min}(K)}$, so the singular values of A span $\sim e^{2\beta \|K\|}$ and the naive product loses all precision within a few tens of slices.

We follow the standard remedy of Loh and Gubernatis [3]: maintain an explicit QR factorisation (Q, R) of the partial product as we left-multiply by each B . Concretely:

1. Initialise $Q = R = \mathbb{K}$ and an accumulator $C = \mathbb{K}$.
2. For $\text{step} = 0, 1, \dots, L_\tau - 1$: set $C \leftarrow B_{(l+\text{step}) \bmod L_\tau} C$. Every `n_stab` steps, refactorise $C Q = Q' R'$ by Householder QR, then update $Q \leftarrow Q'$, $R \leftarrow R' R$, $C \leftarrow \mathbb{K}$.
3. After the last step, the partial product equals QR exactly, with Q orthogonal to working precision.

Then

$$G_\sigma = (\mathbb{K} + QR)^{-1} = (Q(Q^\top + R))^{-1} = (Q^\top + R)^{-1} Q^\top, \quad (13)$$

which is solved by a plain LU on $(Q^\top + R)$. Even though R itself can have a huge dynamical range, the linear system (13) is conditioned by G (whose entries are $O(1)$), and the solve is stable. In our test runs the `max_recompute_drift` reported by the engine – the ℓ^∞ discrepancy between the wrapped G and the freshly recomputed G – stays at machine precision ($\leq 10^{-12}$) for all parameter regimes we have tried up to $\beta U \sim 32$, $N_s = 36$.

8 Observables

For a fixed Hubbard–Stratonovich configuration the system is non-interacting and Wick’s theorem applies exactly. With our convention $G_\sigma(i, j) = \langle c_{i\sigma} c_{j\sigma}^\dagger \rangle$ we have $\langle n_{i\sigma} \rangle = 1 - G_\sigma(i, i)$ and

$$\langle S_i^z S_j^z \rangle = \frac{1}{4} \left[(1 - G_\uparrow(i, i))(1 - G_\uparrow(j, j)) + (1 - G_\downarrow(i, i))(1 - G_\downarrow(j, j)) \right. \quad (14)$$

$$\left. - 2(1 - G_\uparrow(i, i))(1 - G_\downarrow(j, j)) + (\delta_{ij} - G_\uparrow(j, i)) G_\uparrow(i, j) + (\uparrow\leftrightarrow\downarrow) \right]. \quad (15)$$

The implementation evaluates $\langle n \rangle$, the double occupancy $\frac{1}{N_s} \sum_i (1 - G_\uparrow(i, i))(1 - G_\downarrow(i, i))$, $\langle m_z^2 \rangle = (1/N_s^2) \sum_{ij} \langle S_i^z S_j^z \rangle$, the staggered moment $\langle m_{\text{stag}}^2 \rangle = (1/N_s^2) \sum_{ij} (-1)^{a_i + a_j} \langle S_i^z S_j^z \rangle$, and the antiferromagnetic structure factor $S(\pi, \pi) = (1/N_s) \sum_{ij} (-1)^{a_i + a_j} \langle S_i^z S_j^z \rangle$. Errors use the same Flyvbjerg–Petersen logarithmic binning analysis as the SSE code (`qmc::Observable`).

9 Verification

The free-fermion limit ($U = 0$) is the most powerful regression test: the auxiliary field decouples, every configuration produces the same analytical Green’s function, and the simulation must reproduce it to machine precision. Concretely,

$$|G_\sigma^{\text{DQMC}} - (\mathbb{K} + e^{-\beta K})^{-1}|_\infty < 10^{-13} \quad (16)$$

both at initialisation and after a full sweep on the L=8 chain at $\beta = 4$ (`tests/test_dqmc.cpp`).

A second exact identity — $\langle n \rangle = 1$ per configuration on a half-filled bipartite lattice — is reproduced to $\sim 10^{-9}$ on the L=4 chain at $U = 4, \beta = 2$. The reported average sign is exactly +1 in this sign-free regime.

Table 1 summarises a handful of moderate runs that illustrate the expected physics:

System	β	$\langle n \rangle$	$\langle n_\uparrow n_\downarrow \rangle$	$S(\pi, \pi)$	$\langle \text{sign} \rangle$
chain $L = 8, U = 2$	4	1.000	0.169	0.36	1.0
square $4 \times 4, U = 4$	4	1.000	0.124	0.66	1.0
square $6 \times 6, U = 4$	8	1.000	0.120	1.58	1.0

Table 1: Equal-time observables on three half-filled benchmarks. The growth of $S(\pi, \pi)$ with system size and β reflects the expected antiferromagnetic ordering tendency of the half-filled square Hubbard model.

Code map

References

- [1] R. Blankenbecler, D. J. Scalapino, R. L. Sugar, *Monte Carlo calculations of coupled boson–fermion systems. I*, Phys. Rev. D **24**, 2278 (1981).

File	Purpose
<code>include/qmc/linalg.hpp</code>	Dense matrix, GEMM, LU, QR, $\exp(-\Delta\tau K)$, stable G
<code>include/qmc/hubbard.hpp</code>	Hubbard parameters, kinetic matrix
<code>include/qmc/dqmc_engine.hpp</code>	B-builder, sweep, Sherman–Morrison, wrap, recompute
<code>include/qmc/dqmc_measurements.hpp</code>	Equal-time observables (Wick contractions)
<code>apps/qmc_dqmc.cpp</code>	CLI driver, OpenMP parallel replicas
<code>tests/test_dqmc.cpp</code>	Free-fermion + half-filling regression tests
<code>tests/test_linalg.cpp</code>	Linear algebra unit tests

- [2] J. E. Hirsch, *Discrete Hubbard–Stratonovich transformation for fermion lattice models*, Phys. Rev. B **28**, 4059 (1983).
- [3] E. Y. Loh Jr. and J. E. Gubernatis, *Stable numerical simulations of models of interacting electrons in condensed-matter physics*, in *Electronic Phase Transitions*, ed. W. Hanke and Yu. V. Kopayev, Modern Problems in Condensed Matter Sciences **32**, 177 (1992).
- [4] F. F. Assaad and H. G. Evertz, *World-line and Determinantal Quantum Monte Carlo Methods for Spins, Phonons and Electrons*, Lect. Notes Phys. **739**, 277 (2008).